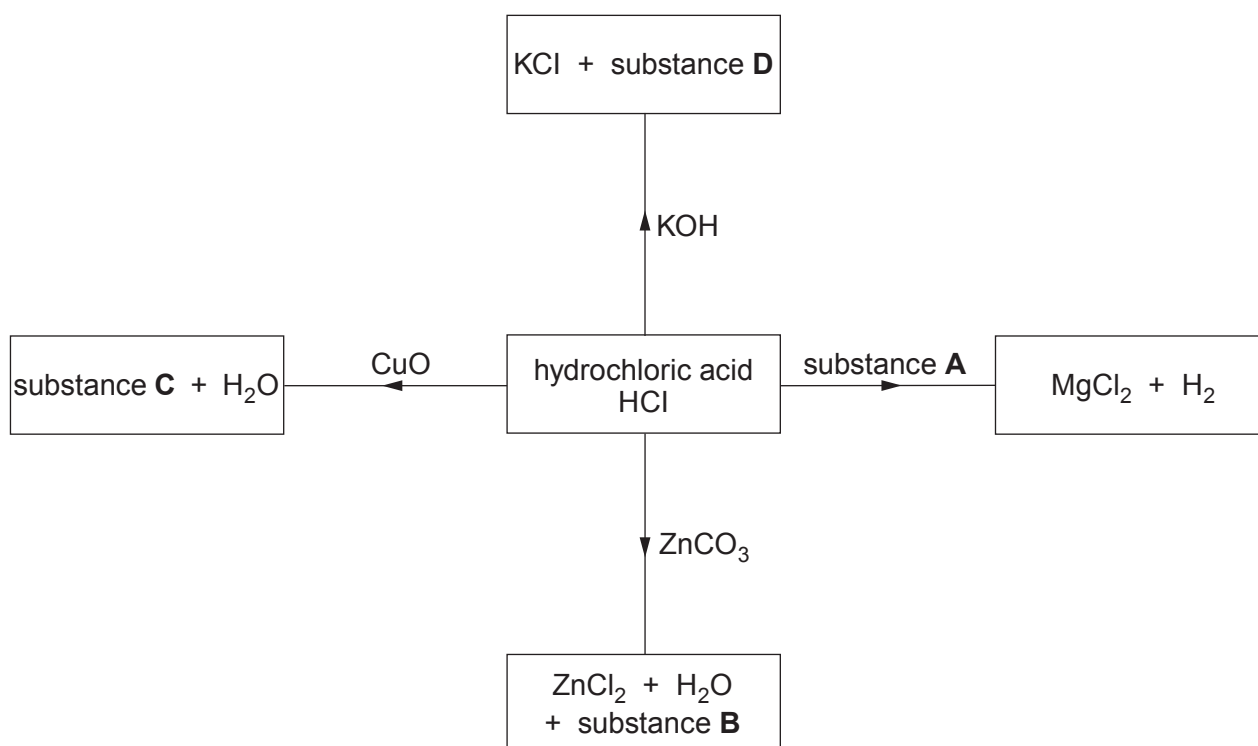


2. The reactions of acids with metals, bases and carbonates are summarised in the following equations.

- acid + metal  $\rightarrow$  salt + hydrogen
- acid + base  $\rightarrow$  salt + water
- acid + carbonate  $\rightarrow$  salt + water + carbon dioxide

(a) The diagram shows some reactions of hydrochloric acid, HCl.



(i) Give the **names** of substances **A** and **B**.

[2]

Substance **A** .....

Substance **B** .....

(ii) Give the **formulae** of substances **C** and **D**.

[2]

Substance **C** .....

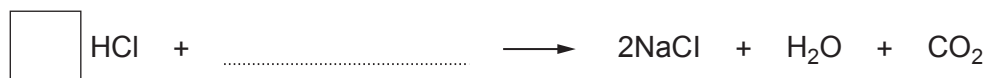
Substance **D** .....



- (b) Complete the equation for the reaction between hydrochloric acid and sodium carbonate by

- writing the formula of sodium carbonate on the dotted line
- putting a number into the box to balance the equation

[2]



- (c) Silver nitrate solution is used to identify the chloride ions present in hydrochloric acid.

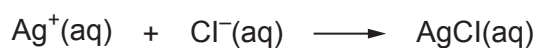
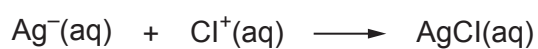
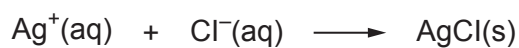
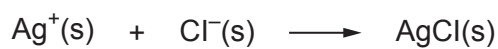
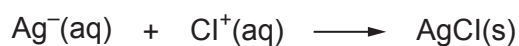
- (i) Give the observation made when silver nitrate solution is added to hydrochloric acid.

[1]

.....

- (ii) Put a tick (✓) in the box next to the correct ionic equation for the reaction between silver nitrate and hydrochloric acid.

[1]

☐☐☐☐☐